

MINISTRY OF ICT SHOWCASE BRIEF

Uganda GovHub API

Secure interoperability, API governance, and sandbox testing for MDA data exchange.

System Description | Professional 5-page presentation PDF | May 2026

Executive positioning

Uganda GovHub API is a working prototype for a government API developer portal, sandbox, and governance console. It demonstrates how Ministries, Departments, and Agencies can publish reusable API services, request access for lawful purposes, test integrations safely, and preserve a traceable record of inter-agency data exchange.

PURPOSE

Show how government APIs can be discovered, governed, tested, and audited from one trusted experience.

AUDIENCE

MoICT leadership, NITA-U, MDA data owners, compliance reviewers, and technical teams.

SCOPE

A live prototype using mock records only. It does not connect to production government systems.

What the system contains

The prototype is a local full-stack application that combines an API catalogue, OpenAPI documentation, sandbox execution, access management, oversight dashboards, and API onboarding in one experience.

API catalogue

Search and filter MDA-owned services by sector, lifecycle, sensitivity, compliance status, sandbox availability, and owning institution.

OpenAPI documentation

Render formal contracts with endpoints, schemas, request/response examples, downloadable assets, and generated code samples.

Sandbox services

Mock endpoints for identity, tax, business registration, driving permit, and composite eligibility scenarios with structured responses.

Governance dashboard

Review approvals, user verification, active keys, access matrix, audit logs, usage analytics, and denied sandbox traffic.

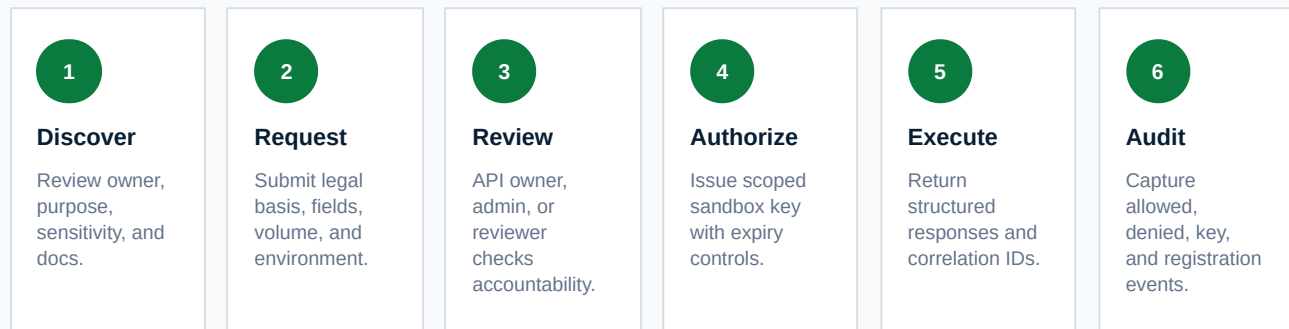
Seeded showcase APIs

OWNER	API PRODUCT	PRIMARY DEMO VALUE	CLASSIFICATION
NIRA	Identity Verification	Verify citizen identity with minimum returned fields.	Restricted
URA	Tax Compliance Status	Check binary tax clearance status for compliance workflows.	Official
URSB	Business Registration Lookup	Validate business status and registry facts.	Public
MoWT	Driving Permit Verification	Confirm permit validity and class authorisation.	Official
MoICT	Service Uganda Composite Eligibility	Show multi-agency eligibility orchestration with correlation IDs.	Restricted

Positioning: GovHub API is not a production replacement for UGHub. It models the governance, documentation, sandbox, and adoption layer around interoperable APIs.

Interoperability as a controlled lifecycle

The system presents API exchange as a policy workflow. Every consumer must state purpose, every owner remains accountable for data access, every credential has scope, and every sandbox transaction has a trace.



Security controls shown

- Cookie-backed sessions and seeded accounts for demo roles.
- Role-based access for developers, API owners, reviewers, and administrators.
- MFA support, TLS/HSTS hooks, encryption-at-rest support, and key hashing.
- Structured denial for missing, expired, revoked, or wrongly scoped keys.

Governance metadata shown

- Owning MDA, technical owner, approval authority, SLA, and contact office.
- Statutory basis, lawful purpose, data minimization, and retention class.
- Personal data categories and security classification for each API.
- Access matrix showing which consumers have approved active channels.

Privacy posture: The demo makes data protection visible before access is granted. It highlights purpose limitation, data minimization, retention, and auditability without exposing live personal data.

A 10-minute demo from discovery to audited exchange

The recommended narrative follows a Ministry of Health service team that needs to verify eligibility for a bundled public service. The presenter shows a complete path without connecting to live systems.

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|----------|-------------------------|--|
| 1 | Open catalogue | Search for identity, tax, business, permit, and composite eligibility APIs. |
| 2 | Inspect API | Show owner, sector, sensitivity, statutory basis, purpose limitation, SLA, and OpenAPI docs. |
| 3 | Request access | Enter business purpose, lawful basis, requested fields, volume tier, and sandbox environment. |
| 4 | Approve request | Switch to platform admin or API owner, approve the request, and generate a scoped sandbox key. |
| 5 | Run sandbox call | Use the approved key, then attempt a denied call to demonstrate policy enforcement. |
| 6 | Review oversight | Open the access matrix, audit logs, and analytics to show active sharing and traceability. |
| 7 | Register API | Validate an OpenAPI specification and capture governance metadata before adding it to the catalogue. |

What changes for MDAs

Service teams

Move from manual verification to reusable, documented API services.

API owners

Control who consumes agency data, for what purpose, and under which approval.

MolCT oversight

See active channels, failed attempts, analytics, and audit evidence in one place.

Developers

Use standard documentation, code samples, sandbox testing, and structured errors before production work.

Prototype readiness with a clear path to pilot validation

React / Vite portal

catalogue, docs, access forms

Express API

routes, auth, policy checks

SQLite MVP store

catalogue, access, audit

OpenAPI assets

contracts and versions

Sandbox services

deterministic mock responses

Current readiness

- Five seeded government-style APIs and ten seeded MDAs for repeatable demonstrations.
- Demo accounts for platform administrator, MDA developer, API owner, and compliance reviewer.
- Documented presenter path and Playwright automation for the ministry demo flow.
- Backend tests cover access control, docs visibility, sandbox policy, auth, versioning, and security features.
- Accessibility checks are available for core demo pages.

Production pilot considerations

- Align roles, catalogue fields, and onboarding workflow with MoICT, NITA-U, and participating MDA operating procedures.
- Decide whether the portal wraps existing UGHub capabilities, operates as an adjacent sandbox, or supports training and onboarding.
- Run DPIA, threat modelling, penetration testing, key-management design, logging policy, and retention review before live records.
- Select two or three low-risk services and define success metrics for turnaround time, reuse, and audit completeness.

Public context reviewed

MoICT Digital Transformation Roadmap 2023/2024-2027/2028: ict.go.ug/wp-content/uploads/2023/08/Digital-Transformation-Roadmap.pdf

NITA-U Integration Service (UGHub): www.nita.go.ug/services/e-government-services/integration-service-ughub

Personal Data Protection Office / Data Protection and Privacy Act No. 9 of 2019: www.pdpo.go.ug/about-us